

1 Gaussian Method

Using Gaussian methods for the variation principle and correlated gaussian method, we define the wavefunction as a sum of gaussians (denoted $|g\rangle$).

$$\psi(\mathbf{r}) = \sum_{i=1}^{n_g} c_i |g_i\rangle = \langle \mathbf{r} | g \rangle.$$

By using this we get the Schrödinger equation as

$$\begin{aligned} H|\psi\rangle &= E|\psi\rangle, \\ \mathcal{H}c &= ENc, \end{aligned}$$

where

$$\begin{aligned} \mathcal{H} &= \{H_{i,j}\} = \langle g_i | \hat{H} | g_j \rangle, \\ \mathcal{N} &= \{N_{i,j}\} = \langle g_i | g_j \rangle. \end{aligned}$$

The correlated gaussian is defined by:

$$\begin{aligned} \langle \mathbf{r} | g \rangle &= \exp \left(- \sum_{i,j=1}^N A_{i,j} \mathbf{r}_i \cdot \mathbf{r}_j + \sum_{i=1}^N \mathbf{s}_i \cdot \mathbf{r}_i \right), \\ &= e^{-\mathbf{r}^T A \mathbf{r} + \mathbf{s}^T \mathbf{r}}. \end{aligned}$$

which gives the notation

$$|g\rangle = |A, s\rangle.$$

The overlap of these gaussians is given by [2]

$$\langle g' | g \rangle = \left(\frac{\pi^N}{\det B} \right) e^{\frac{1}{4} \mathbf{v}^T B^{-1} \mathbf{v}} \equiv M,$$

where $B = A + A'$ and $\mathbf{v} = \mathbf{s} + \mathbf{s}'$.

To find the fouriertransformed correlated gaussian we evaluate the intergral:

$$\begin{aligned} \langle \mathbf{p} | g \rangle &= \int d^{3N} x e^{\mathbf{x}^T \mathbf{p}} e^{-\mathbf{r}^T A \mathbf{r} + \mathbf{s}^T \mathbf{r}}, \\ &= \left(\frac{\pi^N}{\det(A)} \right)^{3/2} e^{\frac{1}{4} (\mathbf{s} - i\mathbf{p})^T A^{-1} (\mathbf{s} - i\mathbf{p})}, \\ &= \left(\frac{\pi^N}{\det(A)} \right)^{3/2} e^{\frac{1}{4} \mathbf{s}^T A^{-1} \mathbf{s}} \cdot e^{-\frac{1}{4} \mathbf{p}^T A^{-1} \mathbf{p}} \cdot e^{-\frac{1}{2} i \mathbf{p}^T A^{-1} \mathbf{s}}, \end{aligned}$$

which gives the fourier transformed gaussian

$$|\tilde{g}\rangle = e^{-\mathbf{p}^T \tilde{A} \mathbf{p} + \tilde{\mathbf{s}}^T \mathbf{p}},$$

where $\tilde{A} = \frac{1}{4}A^{-1}$ and $\tilde{\mathbf{s}} = \frac{-i}{2}A^{-1}\mathbf{s}$. And the $\mathcal{M}$ function is introduced

$$\mathcal{M}(A, \mathbf{s}) = \left(\frac{\pi^N}{\det(A)} \right)^{3/2} e^{\frac{1}{4}\mathbf{s}^T A^{-1} \mathbf{s}},$$

giving us the final solution

$$\langle \mathbf{p} | g \rangle = \mathcal{M}(A, \mathbf{s}) |\tilde{g}\rangle.$$

The overlap in momentum space can be calculated as

$$\begin{aligned} \langle g' | g \rangle &= \int \frac{d^{3N}p}{(2\pi)^{3N}} \langle g' | p \rangle \langle p | g \rangle, \\ &= \left(\frac{\pi^N}{\det A'} \frac{\pi^N}{\det A} \right)^{3/2} \frac{e^{\frac{1}{4}\mathbf{s}'^T A'^{-1} \mathbf{s}'} e^{\frac{1}{4}\mathbf{s}^T A^{-1} \mathbf{s}}}{(2\pi)^{3N}} \langle \tilde{g}' | \tilde{g} \rangle, \\ &= \mathcal{M}(A', \mathbf{s}') \mathcal{M}(A, \mathbf{s}) \frac{\langle \tilde{g}' | \tilde{g} \rangle}{(2\pi)^{3N}}. \end{aligned}$$

We can now use the overlap to calculate the matrix element for the general momentum operator $f(c^T \mathbf{p})$.

$$\begin{aligned} \langle g' | f(c^T \mathbf{p}) | g \rangle &= \int \frac{d^{3N}p}{(2\pi)^{3N}} \langle g' | \mathbf{p} \rangle f(c^T \mathbf{p}) \langle \mathbf{p} | g \rangle, \\ &= \mathcal{M}(A', \mathbf{s}') \mathcal{M}(A, \mathbf{s}) \int \frac{d^{3N}p}{(2\pi)^{3N}} \langle \tilde{g}' | f(c^T \mathbf{x}) | \tilde{g} \rangle, \end{aligned}$$

using equation (??) we continue

$$\begin{aligned} &= \mathcal{M}(A', \mathbf{s}') \mathcal{M}(A, \mathbf{s}) \langle \tilde{g}' | \tilde{g} \rangle \left(\frac{\gamma}{\pi} \right)^{3/2} 2\pi \frac{e^{-\gamma(i\eta)^2}}{i\gamma\eta} \int dx \tilde{f}(x) e^{-\gamma x^2} \sinh(2i\gamma\eta x), \\ &= \langle g' | g \rangle \left(\frac{\gamma}{\pi} \right)^{3/2} 2\pi \frac{e^{\gamma\eta^2}}{\gamma\eta} \int dx \tilde{f}(x) e^{-\gamma x^2} \sin(2\gamma\eta x) \end{aligned}$$

2 Notes

Chatten er sikker på at front faktoren i eq(A.25) burde være $e^{-\gamma\eta^2}$, jeg burde checke udregningen. Coupling term er ikke skrevet for shifted gaussians, så dette burde gøres, chatten har skrevet kode for det.

Litteratur

- [1] Griffiths, D. J. 2018, "Introduction to Quantum Mechanics"
- [2] Dimitri, F. "Analytic matrix elements with shifted correlated Gaussians"
- [3] Navas, S. 2024, "Particle Properties"

https://pdg.lbl.gov/2024/listings/particle_properties.html